

WATER-TANK-FILLING GUARD SYSTEM

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This water-tank-filling guard system for your overhead and ground tanks automatically switches off the pumping system when the overhead tank starts overflowing or the ground tank is empty. This helps to save water and also protect the water pumping system from damage due to dry-run. The system also informs you about different levels of water in the overhead tank and ground tank by giving an audible signal when:

1. Water in the overhead tank reaches the full level.
2. Water in the overhead tank goes below the preset lower bottom level.
3. Water in the ground tank goes below the preset water pumping level.

It can be used in houses, hotels, hospitals, factories, educational institutions, etc.

Circuit description

Fig. 1 shows the circuit of the water-tank-filling guard system for the overhead and ground tanks. The circuit is built around popular seven-transistor Darlington array IC ULN2004, transistors BC558 and BC548, diode 1N4007, voltage regulator 7812 and a few discrete components. To derive the power supply for the circuit, the 230V AC, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 15V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C2 and regulated by IC 7812 (IC2).



The regulated 12V DC so obtained is further filtered by capacitor C1. LED1 acts as a power-on indicator. R11 is a decoupling resistor and C3 is a filter capacitor.

Water-level sensing electrodes L1 through L4 along with CS1 are suspended in the overhead tank to sense different water levels (overflow, full, half and reserve). The +12V DC power supply is connected to common sensing electrode terminals CS1 and CS2.

Working of circuit is simple. As the water level in the overhead tank rises, it bridges electrodes CS1 and

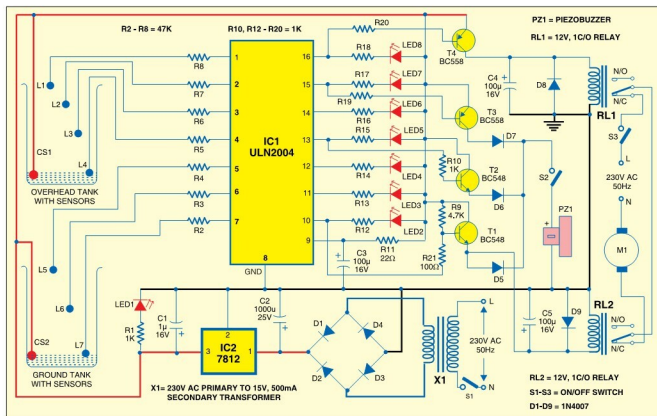


Fig. 1: Circuit for water-tank-filling guard system